

WRRRA-M 0.3 Source Law Filter Selection

Existence Uniqueness Convergence and Stability in the Four Filter Sector

Author	Wonsik Choi
ORCID	0009-0001-4263-9772
Email	janefather@gmail.com
Version	WRRRA-M 0.3-r1
Date	1 October 2026
Status	Conditional selection theorem

Research decision

This note takes the single open assumption left by WRRRA-M 0.2 and turns it into a finite selection problem. The cross-dimensional filter is not added as independent machinery. It is defined as the stable winner of a Source-supplied compatibility table under a fixed Law. The result is exact within a four-filter admissible class and remains conditional on the physical origin of the compatibility values.

Abstract

WRRM 0.3 defines the minimal admissible sector of four one-to-one channel pairings that preserve the sixteen channels and the common color action inherited from version 0.2. Source supplies two scalar compatibility matrices, one for the left block and one for the right block. Law assigns each filter the sum of its selected compatibilities and evolves positive filter weights by a constant-score replicator flow. Two selection gaps completely decide the outcome. The 0.2 cross-dimensional filter is the unique maximizer exactly when both gaps are positive. With positive selection rate $\eta > 0$, the flow converges to it from every interior initial condition, with an explicit exponential bound. The same filter survives entrywise score perturbations of size ρ whenever the smaller gap exceeds four ρ . This establishes existence, conditional uniqueness, convergence, and perturbation stability inside the stated four-filter sector. It does not derive the compatibility matrices from fundamental physics or show that nature gives the required positive gaps.

Result in one sentence

Within the four-filter sector, Source and Law select the WRRM 0.2 cross filter uniquely and stably if and only if both independently fixed selection gaps are positive. Dynamical convergence additionally requires $\eta > 0$ and nonzero initial target support.

Scope and stopping line

Version 0.3 addresses filter selection only and preserves the sixteen-channel algebra of 0.2. Fixed compatibility tables are the score inputs. Known particle assignments may calibrate those tables under the WRRM evaluation rule, provided their provenance is recorded and calibrated selection is distinguished from a calculation using carrier responses fixed before evaluation.

Included in 0.3

- A complete four-filter admissible class for the two binary pairing decisions
- Source compatibility matrices and a fixed scalar score
- A necessary and sufficient two-gap selection criterion
- A closed-form selection flow and convergence bound
- A worst-case perturbation stability margin
- A non-circular evaluation protocol and explicit falsifiers

Reserved for 0.4 or later

- A first-principles calculation of the compatibility matrices
- Proof that the four-filter class is complete in a deeper theory
- Time-dependent scores, stochastic Source noise, or quantum amplitudes
- Three generations, masses, symmetry breaking, gravity, dark matter, cosmology, or consciousness

Continuity from version 0.2

Version 0.2 fixed the input space and proved the consequences of one target pairing. It left the pairing itself unselected. Version 0.3 keeps that input and target unchanged, then compares it with every alternative obtained by independently swapping the left and right singlet assignments.

$$C_{16} = \mathbf{1}_0 \oplus \mathbf{7}_A \oplus \mathbf{7}_B \oplus \mathbf{1}_N$$

$$F_{\times} : C_{16} \longrightarrow (\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\overline{\mathbf{4}}, \mathbf{1}, \mathbf{2})$$

The first filter label records the left assignment and the second records the right assignment. D means the direct left assignment or direct right assignment defined below. X means its swap. The target filter is F-DX: direct on the left and crossed on the right.

$$P_4 = \{F_{DX}, F_{XX}, F_{DD}, F_{XD}\}$$

Fixed WRRR evaluation rule

Stage	Question answered in this note
Verification input	Which 0.2 channels and which established finite-flow identities are held fixed
WRRR-specific transformation	How Source scores candidate pairings and how Law evolves their weights
Output	Which gap signs select the cross filter and how fast the weight converges
Falsification condition	Which independently fixed score or dynamical failure rejects the 0.3 selection claim

The four admissible filters

The minimal candidate class changes only the two binary matchings that 0.2 needs. Every candidate is a channel-preserving bijection. No candidate mixes color components or changes the common SU(3) action. Once the labels A, B, N, and 0 are fixed, the four candidates are distinct.

Filter	Left assignment	Right assignment	Relation to 0.2
F-DX	3-A with 1-A and 3-B with 1-B	anti-3-A with 1-N and anti-3-B with 1-0	Target cross filter
F-XX	3-A with 1-B and 3-B with 1-A	anti-3-A with 1-N and anti-3-B with 1-0	Left swap
F-DD	3-A with 1-A and 3-B with 1-B	anti-3-A with 1-0 and anti-3-B with 1-N	Right swap
F-XD	3-A with 1-B and 3-B with 1-A	anti-3-A with 1-0 and anti-3-B with 1-N	Both swaps

Source compatibility values

Source supplies two real scalar tables. The left table compares each triplet origin with each internal singlet origin. The right table compares each antitriplet origin with the distinguished null and invariant singlets. These values are pairing compatibilities, not linear maps between inequivalent SU(3) representations. Their scalar form therefore does not mix or alter the color action.

$$W_L = (\ell_{ij})_{i,j \in \{A,B\}}, \quad W_R = (r_{ij})_{i \in \{A,B\}, j \in \{N,0\}}$$

The values may later be computed from residue, transition cost, or another upstream invariant. Version 0.3 does not choose that construction. It requires the tables to be fixed before the target filter is evaluated.

Law and the filter score

Law gives each candidate the sum of the four compatibility entries selected by its two matchings. Additive constants common to every entry of one table do not affect the winner, and positive rescaling does not change a gap sign. The target score is shown below.

$$C(F_{DX}) = l_{AA} + l_{BB} + r_{AN} + r_{B0}$$

Filter	Total score
F-DX	$l_{AA} + l_{BB} + r_{AN} + r_{B0}$
F-XX	$l_{AB} + l_{BA} + r_{AN} + r_{B0}$
F-DD	$l_{AA} + l_{BB} + r_{A0} + r_{BN}$
F-XD	$l_{AB} + l_{BA} + r_{A0} + r_{BN}$

Selection gap theorem

Define the left gap as the direct-left score minus the swapped-left score, and the right gap as the crossed-right score minus the direct-right score. Because the candidate set is finite and nonempty, at least one maximizing filter always exists. The target filter is the unique maximizer exactly when both gaps are positive.

$$\Delta_L = (l_{AA} + l_{BB}) - (l_{AB} + l_{BA}), \quad \Delta_R = (r_{AN} + r_{B0}) - (r_{A0} + r_{BN})$$

Proof

Subtract each competing score from the target score. The three differences are the left gap, the right gap, and their sum. If both gaps are positive, all three differences are positive. Conversely, if the target is unique, it must beat the candidate that differs only on the left and the candidate that differs only on the right, so each gap must be positive. A zero gap produces at least a two-way tie. A negative gap selects the corresponding swapped assignment.

$$C_{DX} - C_{XX} = \Delta_L, \quad C_{DX} - C_{DD} = \Delta_R, \quad C_{DX} - C_{XD} = \Delta_L + \Delta_R$$

$$F_x = F_{DX} \text{ is unique} \iff \Delta_L > 0 \text{ and } \Delta_R > 0$$

Gap condition	Selection result
Delta-L > 0 and Delta-R > 0	F-DX is unique
Delta-L = 0 or Delta-R = 0	At least two candidates tie
Delta-L < 0	The left-swapped candidate beats its direct-left counterpart
Delta-R < 0	The right-direct candidate beats its crossed-right counterpart

Selection flow

To turn score ordering into a dynamical selection, Law evolves a positive weight p_F for each filter by the constant-score replicator equation. This standard flow preserves positivity and the unit sum [2]. Here it is used as a defined WRRRA selection law rather than as a biological analogy. The convergence theorem assumes $\eta > 0$. If $\eta = 0$, every weight remains fixed; if $\eta < 0$, interior trajectories favor the lowest scores instead.

$$\dot{p}_F = \eta p_F (C(F) - \bar{C}), \quad \bar{C} = \sum_{G \in \mathcal{F}_4} p_G C(G)$$

For constant scores the flow has the closed solution below. The ratio of any competitor to the target decays exponentially at the corresponding score difference. If $\eta > 0$, the target starts with nonzero weight and both gaps are positive, its weight approaches one.

$$p_F(t) = \frac{p_F(0)e^{\eta C(F)t}}{\sum_{G \in \mathcal{F}_4} p_G(0)e^{\eta C(G)t}}$$

The smallest competing score difference is delta, the smaller of the two gaps. It gives a uniform convergence bound and an explicit time sufficient to reach target weight at least one minus epsilon.

$$1 - p_x(t) \leq \frac{1 - p_x(0)}{p_x(0)} e^{-\eta \delta t}, \quad \delta = \min(\Delta_L, \Delta_R)$$

$$t \geq \frac{1}{\eta \delta} \log\left(\frac{1 - p_x(0)}{p_x(0)\epsilon}\right) \implies p_x(t) \geq 1 - \epsilon$$

Nonzero initial target support is necessary for convergence to that target. Any candidate with exactly zero initial weight remains absent under the flow. For comparison of all four admissible filters, the model assumes positive initial support for each candidate; that stronger model convention is sufficient rather than mathematically necessary for target convergence.

Perturbation stability

Suppose every compatibility entry changes by at most ρ . Each gap contains four signed entries, so its worst-case change is at most four ρ . The target selection therefore survives every such perturbation when the smaller original gap is greater than four ρ . This bound is sufficient and worst-case sharp for independent entry perturbations.

$$\|\widetilde{W} - W\|_{\infty} \leq \rho, \quad \min(\Delta_L, \Delta_R) > 4\rho \implies \widetilde{F}_* = F_x$$

Frozen-input evaluation protocol

For a selection test using inputs fixed before evaluation, use the following order. A calibration run may use known particle assignments, with its provenance recorded separately.

1. Specify the admissible filter class without using the desired charge output.
2. Compute or measure both compatibility tables from Source and Law by a rule fixed in advance.
3. Freeze the tables, the positive selection rate $\eta > 0$, and the initial support before evaluation.
4. Evaluate both gaps and publish their signs, magnitudes, and uncertainty bounds.
5. Apply the flow without retuning and compare the selected filter with F-DX.

Calibrating compatibility values against a known particle assignment is legitimate model construction. The resulting selection is a calibration result. Once the tables are frozen, the score ordering, flow, and stability margin are exact conditional calculations. A selection from carrier responses fixed before checking the target has a different evidential basis. Record which basis is used; calibration itself is not a failure of the model.

Claim ledger

Code	Status	Statement in WRR-M 0.3
E	External input	The 0.2 channel space and the standard constant-score replicator identity
D	Architecture definition	The four-filter class, Source compatibility tables, additive score, and Law flow
T	Exact theorem	Two-gap uniqueness criterion, closed flow solution, convergence bound, and perturbation margin
C	Conditional result	F-DX is uniquely and stably selected when both fixed gaps are positive
H	Unproved hypothesis	Nature supplies the stated candidate class, positive gaps, and nonzero initial support; convergence also assumes $\eta > 0$
O	Outside this version	Upstream score physics, additional filters, generations, masses, gravity, cosmology, and consciousness

Falsification conditions

The 0.3 selection claim fails within its stated scope if any one of the following occurs.

- An independently specified Source and Law give Delta-L less than or equal to zero or Delta-R less than or equal to zero.
- An admissible fifth filter is found with a score at least as large as F-DX, invalidating completeness of the four-filter sector.
- The filter weights do not obey the published ratio law under fixed scores and rate.
- The observed score uncertainty is large enough that the claimed stability margin does not hold.
- Compatibility values or the admissible class are changed after the target outcome is inspected.

Exact boundary

Version 0.3 derives no numerical compatibility entry from fundamental physics. It proves what follows once Source supplies independently fixed scalar compatibilities and Law uses the stated flow. Existence means existence of a maximizer in the finite admissible class, not proof that a physical pre-spacetime system realizes that class. Uniqueness and stability are conditional on positive measured or derived gaps. Convergence requires $\eta > 0$ and positive initial target support; table calibration may be used with its provenance disclosed.

Conclusion

WRRRA-M 0.2 showed what the cross-dimensional filter would produce. Version 0.3 states the minimal conditions under which Source and Law select that filter rather than one of its three pairing alternatives. Two numbers decide the result. Positive left and right gaps make the target unique and protect it against bounded score errors. With $\eta > 0$ and nonzero initial target support, its weight converges to one at a calculable rate. Zero or negative gaps expose failure without interpretive escape. The next task is to fix an upstream compatibility rule and calculate both gaps. Record whether the rule is calibrated against known assignments or supplied by carrier responses fixed before the target check.

Appendix A Compact derivation

Step	Fixed input	WRRRA-M operation	Output
1	The 0.2 sixteen-channel space	Enumerate two binary matchings	Four admissible filters
2	Two Source compatibility tables	Sum selected entries	One score per filter
3	Four scores	Subtract competitors from F-DX	Delta-L Delta-R and their sum
4	Positive gaps, $\eta > 0$	Apply constant-score selection flow	Convergence to F-DX
5	Entrywise error bound ρ	Propagate worst-case gap error	Selection stable if min gap exceeds 4 ρ

$$\Delta_L = (\ell_{AA} + \ell_{BB}) - (\ell_{AB} + \ell_{BA}), \quad \Delta_R = (r_{AN} + r_{BO}) - (r_{AO} + r_{BN})$$

$$F_x = F_{DX} \text{ is unique} \iff \Delta_L > 0 \text{ and } \Delta_R > 0$$

Appendix B Verification identities

The public standalone script `calculations/verify_wrrra_m_0_3.py` checks the three coefficient identities and all 6,561 tables with eight entries in $\{-1,0,1\}$. It verifies the two-gap uniqueness equivalence, exact rational samples of the closed flow and exponential bound, $\eta > 0$, $\eta = 0$, $\eta < 0$, and zero-initial-support cases. It also checks all 256 extreme perturbations of a positive-gap witness and a tie at the sharp stability boundary. Finite enumeration supports the algebraic proof; it does not replace it.

$$p_F(t) = \frac{p_F(0)e^{\eta^{C(F)}t}}{\sum_{G \in \mathcal{F}_4} p_G(0)e^{\eta^{C(G)}t}}$$

Appendix C References

- [1] Wonsik Choi WRRRA-M 0.2 Cross Dimensional Filter Correspondence 2026
<https://github.com/Wonsik-Choi-janefather/wrra-m-0.1/tree/main/paper>
- [2] Peter D. Taylor and Leo B. Jonker Evolutionarily Stable Strategies and Game Dynamics Mathematical Biosciences 40 1978 145 to 156
[https://doi.org/10.1016/0025-5564\(78\)90077-9](https://doi.org/10.1016/0025-5564(78)90077-9)
- [3] Jogesh C. Pati and Abdus Salam Lepton Number as the Fourth Color Physical Review D 10 1974 275 to 289
<https://doi.org/10.1103/PhysRevD.10.275>
- [4] Harald Fritzsch and Peter Minkowski Unified Interactions of Leptons and Hadrons Annals of Physics 93 1975 193 to 266
[https://doi.org/10.1016/0003-4916\(75\)90211-0](https://doi.org/10.1016/0003-4916(75)90211-0)